Python vs. MATLAB Cheat Sheet

*Click each section title to open its Colab notebook (view-only). To run and edit the code, go to **File** → **Save a copy in Drive** and work in your own copy.*

Setup

Task	MATLAB	Python
Import numerics / plotting	*(built in)*	`import numpy as np` `import matplotlib.pyplot as plt`
Workspace info	`whos`	`dir()` (names), `x.__class__`, `type(x)`
Help	`help plot`	`help(plt.plot)`

Scalars, vectors, matrices

Task	MATLAB	Python

Scalar / vector / matrix	`a = 3;`
	`v = [1 2 3];`
	`A = [1 2; 3 4];`
		`a = 3`
		`v = np.array([1, 2, 3])`
		`A = np.array([[1, 2], [3, 4]])`

Column vector	`v = [1; 2; 3];`	`V = np.array([[1], [2], [3]])`
Size / length	`size(A), length(v)`	`A.shape, len(v)`

Zeros / ones / eye	`zeros(3, 2)`
	`ones(2, 3)`
	`eye(3)`
		`np.zeros((3, 2))`
		`np.ones((2, 3))`
		`np.eye(3)`

Linspace / range	`linspace(0, 1, 5)`	`np.linspace(0, 1, 5)`
range step	`1:2:9 -> [1 3 5 7 9]`	`np.arange(1, 10, 2)`
reshape	`reshape(A, m, n)`	`A.reshape(m ,n)`
Concatenate	`[A B]` `[A; B]`	`np.hstcak([A, B])` `np.vstack([A, B])`

Indexing & slicing

Task	MATLAB	Python
Index base	1-based	0-based
Element	`v(1)`	`v[0]`
2D access	`A(i, j)`	`A[i-1, j-1]`
Row / column	`A(i, :)` `A(:, j)`	`A[i-1, ]` `A[:, j-1]`
Slice	`A(2:4, 1:2)`	`A[1:4, 0:2]`

End keyword	`end`	`-1`
Logical mask	`A(A>0)`	`A[A>0]`
Find indices	`[i]=find(v>0)`	`np.where(v>0)`
Fancy indexing	`A([1 3], :)`	`A[[0,2], :]`

Elementwise vs matrix ops

Operation	MATLAB	Python
Elementwise add/sub	`A + B` `A - B`	`A + B` `A - B`
Elementwise mult/div	`A .* B` `A ./ B`	`A * B (elementwise for numpy)` `A / B`
Elementwise power	`A .^ 2`	`A ** 2`
Matrix multiply	`A * B`	`A @ B or np.matmul(A, B)`
Transpose / Hermitian	`A.’` `A’`	`A.T` `A.conj().T`
Dot / inner	`dot(a, b)`	`np.dot(a, b)`

Linear Algebra

Task	MATLAB	Python
Solve Ax = b	`x = A \ b`	`x = np.linalg.solve(A, b)`
Inverse / pseudo inverse	`inv(A)` `pinv(A)`	`np.linalg.inv(A)` `np.linalg.pinv(A)`
Det / trace	`det(A), trace(A)`	`np.lingalg.det(A), np.trace(A)`
Rank	`rank(A)`	`np.linal.matrix_rank(A)`
Eig / SVD	`[V ,D] = eig(A)` `[U, S, V] = svd(A)`	`w, V = np.linalg.eig(A)` `U, S, Vt = np.linalg.svd(A)`
Norms	`norm(A, 2)` `norm(A, ‘fro’)`	`np.linalg.norm(A, 2)` `np.linalg.norm(A, ‘fro’)`

Aggregation & logic

Task	MATLAB	Python
Sum / mean	`sum(v), mean(A, 1)`	`v.sum(), A.mean(axis=0)`
Min / max	`[m, i] = min(v)`	`m = v.min()` `i = v.argmin()`

Any / all	`any(v>0)` `all(v>0)`	`(v>0).any()` `(v>0).all()`
Isnan / isfinite	`isnan(x)` `isfinite(x)`	`np.isnan(x)` `np.isfinite(x)`

Control flow

Task	MATLAB	Python
If / elseif / else	`if a>0 ... elseif ... else ... end`	`if a>0: ...` `elif ... else: ...`
For loop	`for i=1:10, ... end`	`for i in range(1, 11): ...`
While	`while cond, ... end`	`while cond: ...`
Equality / inequality	`==, ~=`	`==, !=`

Functions & scripts

Task	MATLAB	Python
Define function	`function y = f(x), y=x.^2; end`	`def f(x):` `return x**2`
Multiple returns	`[a, b] = g(x)`	`a, b = g(x)`

Anonymous	`f = @(x) x.^2`	`f = lambda x: x**2`
Scripts vs module	`.m file`	`.py file`

Random numbers

Task	MATLAB	Python
Seed	`rng(0)`	`np.random.seed(0)`
Uniform / normal	`rand(3, 2)` `randn(3 ,2)`	`np.random.rand(3, 2)` `np.random.randn(3, 2)`
Permutation	`randperm(n)`	`np.random.permutation(n)`

Plotting (basic)

Task	MATLAB	Python
Simple plot	`plot(x, y)`	`plt.plot(x,y)` `plt.show`

Labels / title	`xlabel('x')`
	`ylabel('y')`
	`title('t')`
		`plt.xlabel('x')`
		`plt.ylabel('y')`
		`plt.title('t')`

Legend	`legend('y')`	`plt.legend(['y']) or`

		`plt.plot(..., label='y')` `plt.legend()`
Subplots	`subplot(2, 1, 1)` `plt(...)`	`fix, ax = plt.subplots(2, 1)` `ax[0].plot(...)`

Images	`imagesc(A)`
	`colorbar`
		`plt.imshow(A, aspect='auto')`
		`plt.colorbar()`
		`plt.show()`

Save figure	`saveas(gfc, 'f.png')`	`plt.save('f.png', dpi=300)`

Data Structures

Concept	MATLAB	Python
Call array	`{1, 2, 3}`	list: `[1, 2, 3]` tuple: `(1, 2, 3)`
Struct	`s.a = 1;`	`dict: s = {'a'=1}`
Table	`table(...)`	`pandas.DataFrame` (via `import panda as pd`)
Strings	`'abc'`	`'abc'` or `"abc"`

Files & I/O

Task	MATLAB	Python
Read text	`fileread('a.txt')`	`open('a.txt').read()`
Load/Save MAT	`load, save`	`scipy.io.loadmat` `scipy.io.savemat`
CVS read/write	`readtable,` `writetable`	`pd.read_csv,` `df.to_csv`